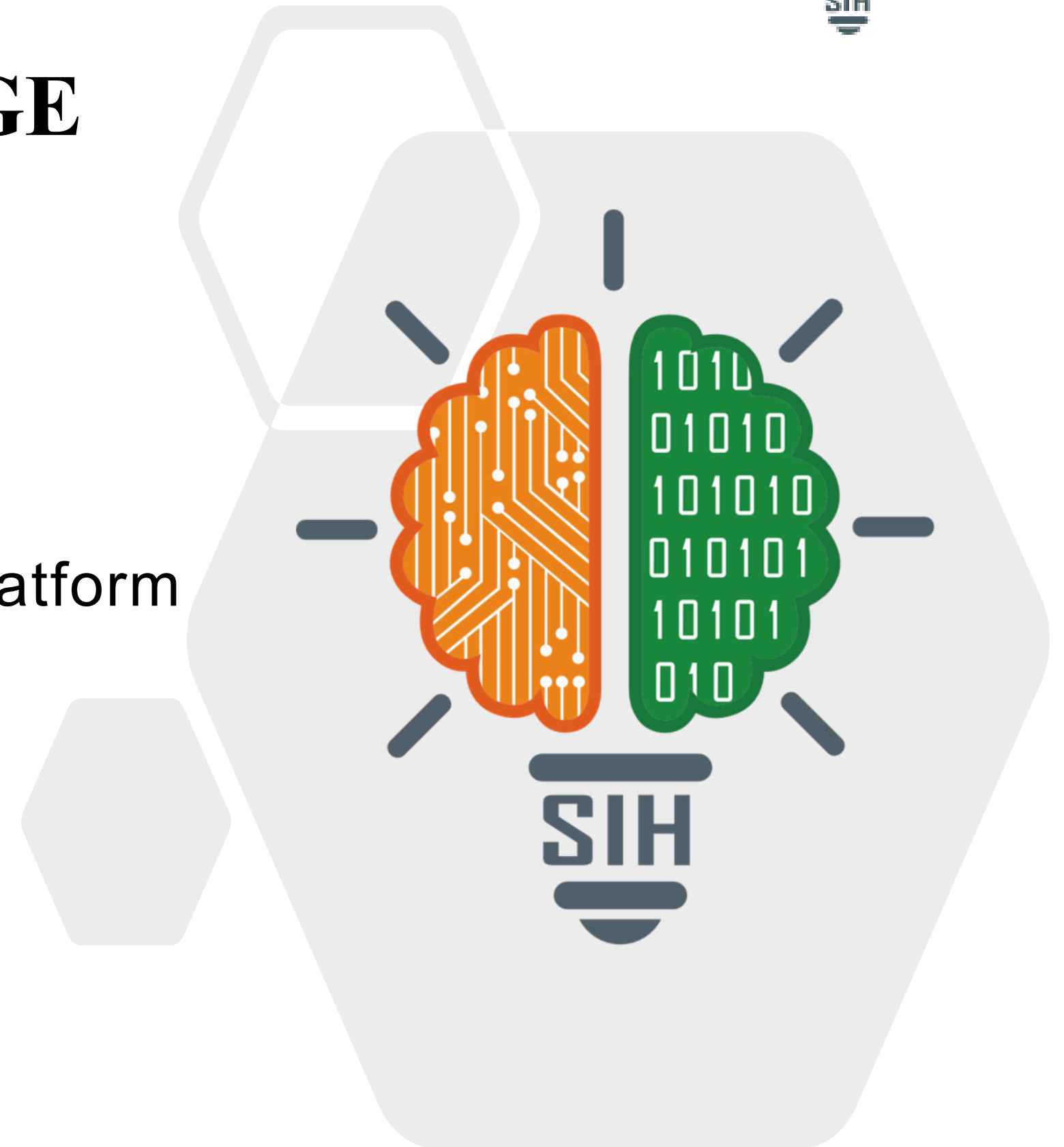


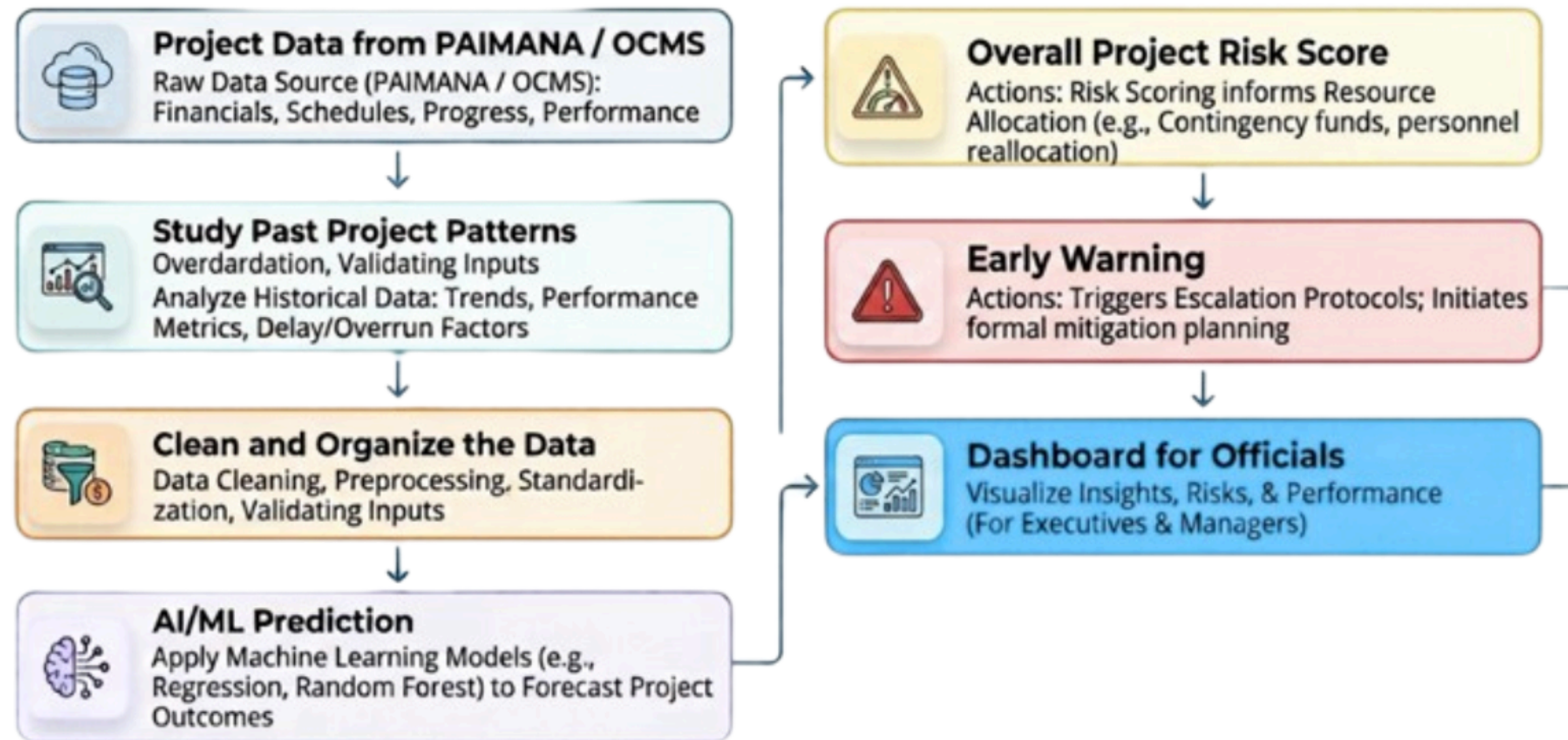
TITLE PAGE

- **Problem Statement ID** - 26103
- **Problem Statement Title**- Use case on web-based integrated project-monitoring platform
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**-
- **Team Name (Registered on portal)**- CritiX



Idea/Solution:

- Our system uses **AI/ML** to enable predictive monitoring of infrastructure projects using **OCMS** and **PAIMANA data**.
- It **predicts potential cost and time overruns** before they occur.
- It **generates a dynamic risk** score based on key project performance indicators.
- It provides **early-warning alerts** for projects at risk of delays or cost escalation.
- It uses **Explainable AI** to identify the key factors behind each risk prediction.
- It enables **benchmarking** and **comparison of projects** across sectors and agencies.



Unique Value Propositions (UVP):

- **Predict & Alert:** Predicts cost/time risks and alerts officials early.
- **Explain & Prioritize:** Scores risk, explains why, and highlights priority projects.
- **Learn & Act:** Learns from past projects and suggests timely actions.

TechStack**Programming & Data Processing**

- Python
- Pandas
- NumPy

AI / ML

- Scikit-learn
- XGBoost / Random Forest
- Statistical prediction models
- Anomaly Detection

Backend

- FastAPI / Flask
- REST APIs

Database

- PostgreSQL / MySQL

Frontend & Visualization

- React.js
- Plotly / Chart.js

LLM

- Open-source LLM
- Retrieval-Augmented
- Generation (RAG)

Deployment

- Docker
- Linux / Open-source infrastructure

1. DATA PREPARATION - Clean • Validate • Handle Missing Data**2. FEATURE ENGINEERING** - Generate project indicators**3. MODEL BUILDING** -Statistical Models + ML Models**4. PREDICTION** -Cost Overrun | Time Delay | Implementation Risk**5. RISK SCORING** - Low ● | Medium ● | High ●**6. EXPLAINABLE AI** - Identify why the project is at risk**7. EARLY WARNING** - Alert the monitoring team**8. DASHBOARD** - Project Health + Risk + Alerts + Trends**Very simple overall flow:**

Analysis of the feasibility of the idea

Technical: Uses PAIMANA data + AI/ML.

Financial: Uses low-cost open-source tools.

Market: Useful for MoSPI and government departments.

Operational: Simple dashboard for easy monitoring.

Scalable: Can cover more projects and sectors.

1

2

3

Strategies for overcoming these challenges

Data: Clean and verify data.

Models: Test and select the best model.

Accuracy: Update models regularly.

Trust: Show reasons behind predictions.

Alerts: Use proper risk levels.

Human + AI: Officials make the final decision

Potential challenges and risks

Technical: Incomplete or incorrect data.

Financial: Higher cost for large-scale data.

Market: User trust in AI predictions.

Operational: Need regular data updates.

AI: Wrong predictions or false alerts.

Potential Impact on Target Audience:

- **MoSPI / IPMD:** Identify high-risk projects early.
- **Monitoring Officers:** Get early alerts and risk reasons.
- **Ministries / Departments:** Track project health and take timely action.
- **Decision Makers:** Compare projects and prioritise resources.
- **Project Teams:** Address risks before they become major issues.
- **Overall:** Shift from reactive monitoring to proactive action.

Benefits of the Solution:

- **Social:** Faster delivery of public infrastructure.
- **Economic:** Early detection of cost increases and better use of public funds.
- **Governance:** Faster, data-driven project monitoring.
- **Operational:** Less manual effort in identifying high-risk projects.
- **Environmental:** Reduce resource wastage from prolonged delays and rework.
- **Long-Term:** Scalable monitoring across sectors and projects.

**1. MoSPI – Infrastructure & Project Monitoring Division (IPMD)**

PAIMANA Project Monitoring Platform

<https://ipm.mospi.gov.in/AboutUs/AboutIPMD>

**2. PAIMANA Public Dashboard**

Project cost, expenditure, progress and sector-wise project information.

<https://ipm.mospi.gov.in/Home/PublicDashboard>

**3. New OCMS – PAIMANA**

Project monitoring, standardized metrics and time/cost-overrun monitoring.

<https://ipm.mospi.gov.in/AboutUs/AboutOCMS>

**4. OCMS – MoSPI FAQ**

Background and role of OCMS in Central Sector Infrastructure Project monitoring.

<https://ipm.mospi.gov.in/FAQ/Index>

**5. Machine Learning & Model Evaluation**

Scikit-learn documentation — model selection, cross-validation and prediction evaluation.

<https://scikit-learn.org/stable/>

XGBoost**6. XGBoost**

Gradient-boosting methods for machine-learning based prediction.

<https://xgboost.readthedocs.io/en/stable/>

7. Explainable AI – SHAP

Used to understand which features contribute to an ML prediction.

https://shap.readthedocs.io/en/latest/example_notebooks/overviews/An%20introduction%20to%20explainable%20AI%20with%20Shapley%20values.html

8. FastAPI

For building the backend APIs connecting the prediction system with the dashboard.

<https://fastapi.tiangolo.com/>